

CS5037 – Data Visualization

Interactive Data Storytelling with Tableau

NOAA Weather Station Data – Annapolis, MD (2010–2025)

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Metric	Value
Total Records	192
Avg Temperature	58.5 °F
Max Temperature	91.1 °F
Min Temperature	19.5 °F
Avg Precipitation	3.58 in
Avg Wind Speed	6.8 mph

1. Dataset Overview

The dataset used in this project comes from NOAA and contains monthly weather observations from Annapolis, Maryland between 2010 and 2025. It includes temperature, precipitation, and wind data, which makes it useful for analyzing seasonal patterns and long-term trends. Overall, the dataset is clean and well-structured for visualization purposes.

2. Data Cleaning and Preparation

Before creating visualizations, I cleaned the dataset by removing missing values and formatting the date column. I also created new fields such as temperature range, season, and categories for temperature and precipitation. These transformations made it easier to analyze patterns in Tableau.

3. Visualizations and Findings

The time-series chart shows a clear seasonal pattern, with high temperatures in summer and low temperatures in winter. Precipitation appears fairly consistent throughout the year, although August tends to be slightly wetter. The scatter plot suggests a weak relationship between temperature and precipitation, while the box plot highlights variability across seasons.

4. Key Insights

One interesting observation is that temperature patterns remain relatively stable across the years. Although there are slight variations, no major long-term trend is clearly visible. Another insight is that spring shows the highest variability in temperature range, which reflects changing weather conditions.

5. Use of Tableau

Tableau made it easy to build interactive dashboards and quickly explore different visualizations. The filtering and linking features helped in analyzing data from multiple perspectives without writing code.